Madhav Anand Menon

+1 (646) 938-7079 | madhavanandmenon@gmail.com | linkedin.com/in/madhav-anand-menon/ | github.com/MadhavMenon10

Education

University of Illinois Urbana-Champaign — Expected May 2028

B.S. Computer Science and Physics; Minor in Mathematics (James Scholar Honors) — **GPA: 3.97/4.0**

- Yee Seung Ng Award; Harold and Ruth Hayward Scholarship; Dean's List; Tau Beta Pi Honors Society
- Relevant Coursework: Honors Data Structures and Algorithms, Computer Architecture, Honors Linear Algebra, Probability Theory, Partial Differential Equations (Upcoming: Operating Systems, Numerical Methods)

Experience

Valeo — Jun. 2026 - Aug. 2026

Software Engineer Intern — Chennai, TN, India

- Engineered a 4-state Extended Kalman Filter in Python fusing gyro, accelerometer, and GPS-velocity data, cutting IMU pitch error by 37% against a survey-grade INS
- Diagnosed a prior filter's failure across a 9,000-sample log, proving that centripetal acceleration corrupted attitude 42 sec before any trigger fired.
- Tuned sensor-fusion gating against a measured 0.91 m/s2 noise floor, cutting false-rejection duty cycle by 60% toward a sub-10% stability target

Disruption Lab — Feb. 2026 - Present

Software Engineer — Champaign, IL, USA

- Built a 5-stage Canvas-to-PostgreSQL RAG pipeline supporting 3 file formats (PDF/DOCX/TXT) with 1024-dim pgvector embeddings and sliding-window chunking via AWS Bedrock
- Designed a skill extractor with NACE framework alignment, returning deduplicated skill taxonomies across 100+ courses per API call, reducing manual tagging effort by 98%
- Shipped a Python FastAPI backend with custom token-based authentication and dual-mode search (full-text PostgreSQL and pgvector cosine similarity) supporting 120+ concurrent users

Algoverse AI — Sep. 2025 - Jun. 2026

LLM Research Intern — San Francisco, CA, USA (Remote)

- Engineered DeployStega, an LLM steganographic communication framework using token-binning, achieving 90%+ word inclusion fidelity validated against a 13M-event corpus spanning 695K+ real GitHub users
- Derived statistical priors from GH Archive to constrain synthetic trace generation within an empirically normal behavioral envelope
- Evaluated detection resistance against 5 adversarial classifiers (LR, RF, SVM, TS-CNN, BERT-base) on 4,000 synthetic users, achieving sub-1.173 population-level indistinguishability

DigiAlert — Jun. 2025 - Aug. 2025

Software Engineer Intern — Chennai, TN, India

- Designed a Python boto3 CloudTrail pipeline sustaining gap-free ingestion under AWS's 2 req/s LookupEvents throttle, using 5-min-overlap incremental windowing with EventId deduplication and 6-retry backoff
- Built a Python Flask-based AWS CSPM dashboard aggregating security metrics across 4 AWS services with CIS/NIST aligned risk classification and GeoLite2-geolocated logins, cutting CSPM tooling costs by 45%
- Shipped via nginx/gunicorn with Chart.js visualizations, dual-tier TTL cache invalidation (10-min memory / 60-min disk), and non-blocking single-flight refresh preventing overlapping AWS fetches

Projects

CUDA-Accelerated Order Book Reconstructor | *CUDA, C++, CUB/Thrust, Nsight Systems*

- Built a CUDA order-book engine end-to-end with a custom ITCH 5.0 binary decoder, per-symbol stream compaction, and warp-per-symbol reconstruction, processing 305M+ NASDAQ messages at 4.4M msgs/sec across 9,000 symbols on an H100
- Swept 10,000 backtest configurations in parallel, compressing 6.5 hours of sequential backtesting into 83.8s for a 279× speedup and 156× end-to-end over a single-threaded C++ baseline on identical hardware
- Engineered a from-scratch GPU open-addressing hash table sustaining 99.99997% reconstruction accuracy, and eliminated 27 GB tick buffer via atomic-claim allocation over pinned SoA buffers

GPU Visualizer | *TypeScript, CSS, HTML*

- Built a GPU die visualiser spanning 8 GPUs from NVIDIA, AMD, and Intel that renders 1,807 clickable blocks nested 6 levels deep, using a typed block schema and recursive SVG renderer in React and TypeScript
- Documented 96 on-die components and 380 spec rows drawn from 9 vendor whitepapers and ISA guides, mapping each block to its function, pipeline stage, and factory fuse-off state
- Cut re-renders on a 361-node die and held a 300x zoom range (0.4x to 120x) smooth under pan and hover by quantising zoom into level-of-detail tiers and memoising the render tree; auto-deployed via GitHub Actions

Skills

Languages: C++; C; Python; TypeScript; SQL; Verilog; MIPS Assembly
Libraries and Frameworks: CUDA; PyTorch; FastAPI; Flask; pandas; scikit-learn; NumPy; Matplotlib
Tools: NVIDIA Nsight; Linux; AWS; Azure; GCP; Git; TeX/LaTeX; Docker; Valgrind, Mathematica, MATLAB
Other Activities: CS 124 Project Manager; CodePath; TEDx Speaker; ACM Corporate Team Member